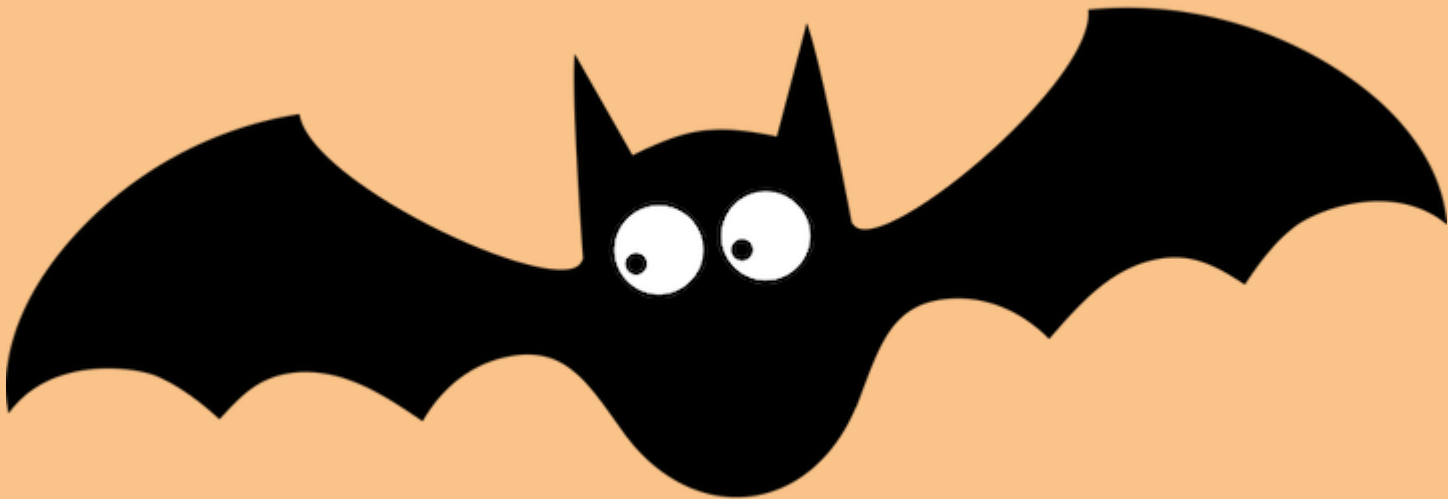




Gradient Boosting

Son Nguyen

Boosting means "Learning from mistakes"

↙
Ada Boost

Increase the weights of the mistakes to fix them in the next rounds

↘
Gradient Boosting

change the target variable by the errors .

(*) Let's predict the temp. tomorrow given the temp. today

L : learning rate
 ($L=1$ in this case
 but in practice
 $L \approx 0.1$)

original
data

X	Y
70	75
60	50
80	90
50	75
100	120

train
M1

X	Y	M1 predicts	Error 1 $Y - M1 \text{ predict}$
70	75	80.2	-5
60	50	66.2	-10
80	90	100.2	-10
50	75	66.2	15
100	120	90.2	30

X	Y_2
70	-5
60	-10
80	-10
50	15
100	30

Train
M2

X	Y_2	M2 predicts	Error 2
70	-5	-4.2	-1
60	-10	-15.2	5
80	-10	-7.2	-3
50	15	12.2	3
100	30	20.2	10

Data for
the third
model

X	Y_3
70	-1
60	5
80	-3
50	3
100	10

train
M3

X	Y_3	M3 predict
70	-1	-1.2
60	5	4.2
80	-3	-3.2
50	3	3.2
100	10	9.2

X	Y	m_1 ↓ M1 predicts	m_2 ↓ M2 predic	m_3 ↓ M3 predict	Final prediction $= m_1 + m_2 + m_3$
70	75	80	-4	-1	$80 - 4 - 1 = 75$
60	50	60	-15	4	$60 - 15 + 4 = 49$
80	90	100	-7	-3	$100 - 7 - 3 = 90$
50	75	60	12	3	$60 + 12 + 3 = 75$
100	120	90	20	9	$90 + 20 + 9 = 119$

⊛ So far, the main hyper-parameter of GB is the number of models in the sequence.

⊛ Learning Rate (shrinkage)

Evaluating a regression model

(true) y	(predicted) $\hat{y}$	Error $(y - \hat{y})$	Squared error $(y - \hat{y})^2$	$ y - \hat{y} $
80	79	1	1	1
100	105	-5	25	5
120	100	20	400	20

① Sum Squared Error (SSE) = $1 + 25 + 400 = 426$

$$SSE = \sum (y - \hat{y})^2$$

② Mean Squared error : (MSE) = $\frac{\sum (y - \hat{y})^2}{3} = \frac{426}{3} = 142$

③ Mean Absolute Error (MAE)

$$MAE = \frac{\sum |y - \hat{y}|}{3} = \frac{26}{3} = 8.67$$

Classification: Baseline model is "guessing" or tossing a coin then make prediction.

Regression: Baseline model is predicting using the average. ($\hat{y}^* = \bar{y}$)

(true) y	$\hat{y}^*$	Error ($y - \hat{y}^*$)	Squared error	$ y - \hat{y}^* $	$\hat{y}_2$
80	100	-20	400	20	1000
100	100	0	0	0	2000
120	100	20	400	20	3000

↑
worse than $\hat{y}$

$$\bar{y} = 100$$

$$SSE^* = 400 + 0 + 400 = 800$$

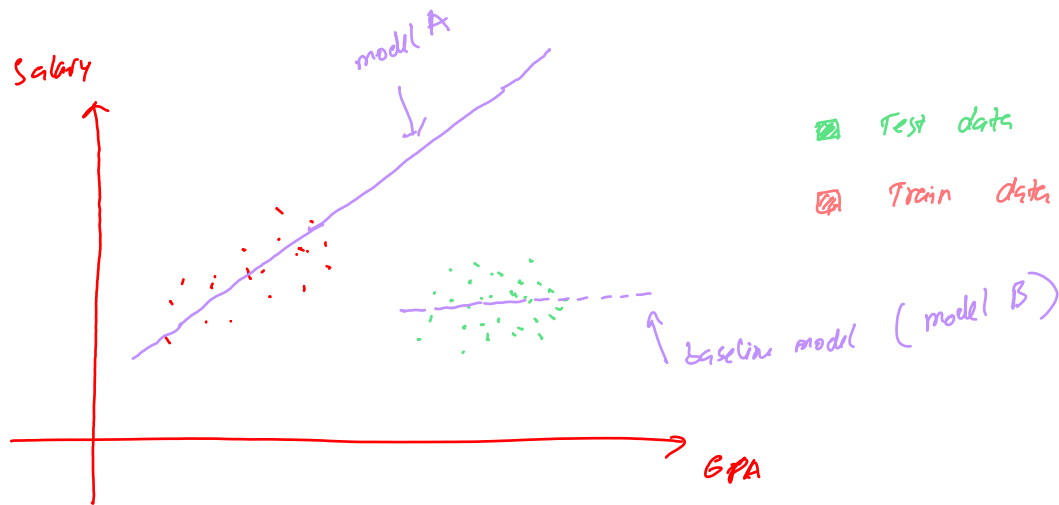
↑
SSE of the
baseline model

$$(*) \quad R\text{-Squared} = 1 - \frac{SSE}{SSE^*} = 1 - \frac{426}{800} = .4675$$

R-squared

- 1 if the model predict perfectly
- 0 if the model is as good as the baseline model (predicting by the average)
- (-) if the model is worse than the average.

↓
if it happens, it often on the **test data**



Model A : Training R-squared : 0.85

Testing R-squared : negative b/c it is worse than model B

